

Standby Diesel Generator — Emergency Power

Project Helios Data Centre · Critical & Life-Safety Loads

Document No. DB-GEN-001 | Rev A | Issued for Tender

1. Scope

This section governs the standby diesel generating set(s) providing emergency power to the critical and life-safety loads of the Helios data centre on loss of utility supply. Where a vendor proposal differs from any requirement below, it shall be recorded as a deviation and is non-conforming unless approved in writing by the Engineer.

2. Performance Requirements

Ref	Parameter	Requirement
2.1	Standby rated power	≥ 2000 kW at site conditions
2.2	Output voltage / frequency	11 kV, 50 Hz , 3-phase
2.3	Emissions compliance	EPA Tier 4 Final (or EU Stage V) certified
2.4	On-site fuel autonomy	≥ 24 hours at 100% rated load without refuelling
2.5	Start & load acceptance	Reach full rated load within 10 seconds of a start signal
2.6	Redundancy	N+1 generating capacity
2.7	Acoustic enclosure	≤ 85 dB(A) at 1 m
2.8	Governor	Electronic isochronous, ≤ ±0.25% steady-state

3. Notes to Tenderer

- Emissions (2.3):** the site requires Tier 4 Final / Stage V certification. Lower tiers (Tier 2, Stage IIIA) require aftertreatment or a formal variance and shall be treated as non-conforming.
- Fuel (2.4):** state usable on-site storage at 100% load. A sub-base "belly" tank sized for only a few hours does not satisfy the 24-hour requirement.
- Redundancy (2.6):** a single set does not satisfy N+1; provide a parallel set or an installed standby unit.